

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1. Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Consider the following sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0, \frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

- (i) Show $X_n \rightarrow X$ almost surely (P).
- (ii) Does $X_n \rightarrow X$ in probability?
- (iii) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$. (7+7+(7+7) = 28 points)

2. Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e. $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

- (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

- (b) Prove: $\sqrt{1 + (E(X))^2} \leq E\sqrt{1 + X^2} \leq 1 + E(|X|)$.

- (c) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

$$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f \cdot g \quad \text{a.e. } (m^2)$$

- (d) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.
- (e) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e. (m), f and g are as in (c).
- (f) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ?

(8+8+8+8+8+(5+4) = 49 points)

3. (i) Show that a *Normal*(0, 1) random variable exists:

In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, such that its c.d.f. is given by Φ where

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du, \quad x \in \mathbb{R}.$$

- (ii) Compute the following Lebesgue integral (Clearly stating any result used)

$$\int X(\omega) dP(\omega).$$

(iii) For each $n \geq 1$, let μ^n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Under what conditions can we say that there is a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $P(A \times \mathbb{R}^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty), P)$ such that the joint distribution of $(X_1, \dots, X_n)$ is given by μ^n , for all $n \geq 1$.

(iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(0, 1)$ random variables. (4×7= 28 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$

1. Solution:

- (i) $\forall \omega \in (0, 1), X_n(\omega) = 0$ for large n since $\frac{1}{n^2} < \omega$ for $n \geq \frac{1}{\sqrt{\omega}}$ if $\omega > 0$. Also we have $X_n(0) = X_n(1) = 0, \forall n \geq 1$. So $X_n(\omega) \rightarrow X(\omega), \forall \omega \in [0, 1]$, i.e. $X_n \rightarrow X$ a.e. (P) .
- (ii) $m(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = m((0, \frac{1}{n^2})) = \frac{1}{n^2} \rightarrow 0$ as $n \rightarrow \infty, \forall \epsilon > 0$. So $X_n \xrightarrow{P} X$.
- (iii) $(E|X_n - X|^p)^{\frac{1}{p} \wedge 1} = (\int |X_n - X|^p dP)^{\frac{1}{p} \wedge 1} = (n^p \cdot m((0, \frac{1}{n^2})))^{\frac{1}{p} \wedge 1} = n^{\frac{p-2}{p \vee 1}} \rightarrow 0$ as $n \rightarrow \infty$ iff $0 < p < 2$. Hence $X_n \rightarrow X$ in L^p for $0 < p < 2$, but this convergence doesn't hold for $2 \leq p < \infty$.

2. Solution:

(a) Note that

$$\begin{aligned} \text{Var}(X + Y) &= E[(X - EX) + (Y - EY)]^2 \\ &= \text{Var}(X) + \text{Var}(Y) + 2E[(X - EX)(Y - EY)] \\ &= \text{Var}(X) + \text{Var}(Y) + 2[E(XY) - EXEY] \end{aligned}$$

Now $E|XY| \leq \sqrt{EX^2EY^2} < \infty$ by Cauchy-Schwarz inequality. So $XY \in L^1(\Omega, \mathcal{F}, P)$. Then, by using Fubini theorem,

$$\begin{aligned} E(XY) &= \int_{\Omega} X(\omega)Y(\omega)dP(\omega) \\ &= \int_{\mathbb{R}^2} xy d(P_X \times P_Y)(x, y) \\ &= \int_{\mathbb{R}} x dP_X(x) \int_{\mathbb{R}} y dP_Y(y) \\ &= \int_{\Omega} X(\omega) dP_X(\omega) \int_{\Omega} Y(\omega) dP_Y(\omega) \\ &= EXEY \end{aligned}$$

Hence $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

- (b) Let $\phi(x) = \sqrt{1 + x^2}$, then $\phi(x)$ is convex, so $\sqrt{1 + (EX)^2} = \phi(EX) \leq E\phi(X) = E\sqrt{1 + X^2}$ given that $EX < \infty$. When $EX = \infty$, this is automatically true. As $1 + X^2 \leq (1 + |X|)^2$, so $E\sqrt{1 + X^2} \leq E(1 + |X|) = 1 + E|X|$.
- (c) From $\frac{dP_X}{dm} = f, \frac{dP_Y}{dm} = g$, we have

$$\begin{aligned} P_{(X,Y)}(A_1 \times A_2) &= (P_X \times P_Y)(A_1 \times A_2) \text{ (by definition of product measures)} \\ &= P_X(A_1)P_Y(A_2) \text{ (by definition of RN derivative)} \\ &= \left(\int I_{A_1} f dm \right) \left(\int I_{A_2} g dm \right) \text{ (by Tonelli's theorem since } f, g \geq 0) \\ &= \int I_{A_1}(x) I_{A_2}(y) f(x)g(y) dm(x)dm(y) \\ &= \int_{A_1 \times A_2} f(x)g(y) dm^2(x, y) \end{aligned}$$

← should this be due to independence

Since on rectangles $A_1 \times A_2$ on $\mathbb{R}^2$,

$$P_{(X,Y)}(A_1 \times A_2) = \int_{A_1 \times A_2} f(x)g(y) dm^2(x, y),$$

By uniqueness of Caratheodory extension theorem, we have

$$P_{(X,Y)}(A) = \int_A fg dm^2, \quad \forall A \in \mathcal{B}(\mathbb{R}^2)$$

Note: another argument is the following

$$\begin{aligned} (P_X \times P_Y)(A) &= \int P_Y(A_{1x}) dP_X(x) \\ &= \int \left[\int I_{A_{1x}} dP_Y(y) \right] dP_X(x) \\ &= \int \left[\int I_{A_{1x}} g(y) dm(y) \right] dP_X(x) \\ &= \int \int I_{A_{1x}}(y) g(y) dm(y) f(x) dm(x) \\ &= \int \int I_A(x, y) f(x) g(y) dm^2(x, y) \end{aligned}$$

[More rigorous proof is: define $\mu(A) = \int_A fg dm^2$, $\forall A \in \mathcal{B}(\mathbb{R}^2)$, then $\mu = P_{(X,Y)}$ on $\mathcal{C} = \{A \times B : A, B \in \mathcal{B}(\mathbb{R}^2)\}$, and it is a finite measure on $\mathcal{C}$. Hence by uniqueness theorem, $\mu = P_{(X,Y)}$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$.]

From this, it is clear that $P_{(X,Y)} \ll m^2$ and $\frac{dP_{(X,Y)}}{dm^2} = f \cdot g$ a.e. (m^2).

(d) Note that $\sup_{x \in \mathbb{R}} |f(x)| = M < \infty$ and $(f * g)(x) = \int f(x-u)g(u) du$.

If $x_n \rightarrow x$, since f is continuous and $\phi(x) = x - u$ is continuous for fixed u , then $K_n(u) = f(x_n - u) \rightarrow f(x - u)$. So for all $u \in \mathbb{R}$,

$$K_n^x(u) \triangleq f(x_n - u)g(u) \rightarrow K^x(u) \triangleq f(x - u)g(u)$$

Then $|K_n^x(u)| \leq Mg(u)$ and $g \in L_1$. By DCT, we have

$$\int K_n^x(u) du \rightarrow \int K^x(u) du$$

as $n \rightarrow \infty$, i.e. $(f * g)(x_n) = \int f(x_n - u)g(u) du \rightarrow \int f(x - u)g(u) du = (f * g)(x)$. So $\forall x_n \rightarrow x$, $(f * g)(x_n) \rightarrow (f * g)(x)$, i.e. $f * g$ is continuous.

Also note that for all x ,

$$\int |f(x - u)g(u)| du \leq M \int |g(u)| du \equiv M' < \infty.$$

So for all x ,

$$|(f * g)(x)| \leq \int |f(x - u)g(u)| du \leq M'$$

Hence $f * g$ is bounded.

- (e) If $P_X \ll m, P_Y \ll m$, then we have $f = \frac{dP_X}{dm}, g = \frac{dP_Y}{dm}$ by the definition of Radon-Nikodym derivatives, the following holds for $A \in \mathcal{B}(\mathbb{R})$ (proving the claim):

$$\begin{aligned}
(P_X * P_Y)(A) &= \int \int I_A(x+y) dP_X(x) dP_Y(y) \\
&= \int \left(\int I_A(x+y) f(x) dm(x) \right) dP_Y(y) \quad (\text{by change of variable}) \\
&= \int \left(\int I_A(u) f(u-y) dm(u) \right) dP_Y(y) \quad (\text{by Fubini's theorem}) \\
&= \int I_A(u) \left(\int f(u-y) dP_Y(y) \right) dm(u) \\
&= \int I_A(u) \left(\int f(u-y) g(y) dm(y) \right) dm(u) \\
&= \int I_A(u) (f * g)(u) dm(u) \\
&= \int_A (f * g)(u) dm(u)
\end{aligned}$$

Hence by the uniqueness of the RN derivatives, $\frac{\lambda}{dm} = f * g$ a.e. (m).

- (f) It says Z has the same law as $X+Y$. Part (d) says if X, Y has densities f, g respectively, then Z will have density $f * g$.

3. Solution:

- (i) Define $\mathcal{C} = \{(a, b], (b, \infty) : -\infty \leq a, b < \infty\}$ and $\mu_\Phi((a, b]) = \Phi(b) - \Phi(a)$, $\mu_\Phi((b, \infty)) = 1 - \Phi(b)$. This defines a measure on the semi-algebra $\mathcal{C}$. Caratheodory extension theorem states that there exists an extension μ^* of μ_Φ to $(\mathbb{R}, \mathcal{M})$, and $\mathcal{C} \subset \mathcal{M}$, where

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu_\Phi(A_n) : \{A_n\} \subset \mathcal{C}, A \subset \bigcup_{n=1}^{\infty} A_n \right\}, \quad A \subset \mathbb{R},$$

and $\mathcal{M}$ is the set of μ^* -measurable sets, i.e. $\mathcal{M} = \{A \subset \mathbb{R} : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c), \forall E \subset \mathbb{R}\}$. Since $\mathcal{C} \subset \mathcal{M}$, and we know that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$, then we have $\mathcal{B}(\mathbb{R}) \subset \mathcal{M}$. Hence, we get an extension of μ_Φ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ by restricting μ^* to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We will denote this measure as μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Define $(\Omega, \mathcal{F}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$. Define $X : \Omega \rightarrow \mathbb{R}$ as $X(\omega) = \omega$. Clearly, it is measurable and μ is a probability (and hence X is a random variable). Also, the law of X is $P_X(A) = P(X^{-1}(A)) = \mu(A)$ for all $A \in \mathcal{B}(\mathbb{R})$. $F_X(x) = P(X \leq x) = \mu((-\infty, x]) = \Phi(x)$, for all $x \in \mathbb{R}$.

- (ii) Here we use the following change of variable formula for Lebesgue integrals: If X is a measurable transformation from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then $X \in L^1(\Omega, \mathcal{F}, P)$ iff $I \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$, where $I(x) = x$, for $x \in \mathbb{R}$, and in that case,

$$\int X dP = \int_{\mathbb{R}} I(x) dP_X.$$

From the definition of P_X in the problem and relationship between Riemann and Lebesgue integrals, we know that the right side of the above equation is

$$\int_{\mathbb{R}} I(x) dP_X = \int_{\mathbb{R}} I(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dm(x) = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx,$$

where the last expression is a usual Riemann integral. We know from elementary calculus that this integral is equal to 0.

- (iii) From Kolmogorov's Consistency theorem, only condition we need here is that $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for all $A \in \mathcal{B}(\mathbb{R}^n)$ for all $n \geq 1$. For the sequence of random variables, define for $\omega = (\omega_1, \omega_2, \dots) \in \mathbb{R}^\infty$, $X_n(\omega) = \omega_n$, $n \geq 1$. Then the joint law of $(X_1, \dots, X_n)$ is given by

$$P_{(X_1, \dots, X_n)}(A) = P((X_1, \dots, X_n) \in A) = P(A \times \mathbb{R}^\infty) = \mu^n(A), \text{ for } \forall A \in \mathcal{B}(\mathbb{R}^n), n \geq 1.$$

- (iv) In part (i), define $\mu^{(1)} = \mu \equiv N(0, 1)$ and $\mu^{(n+1)} = \mu^{(n)} \times \mu$, for $n \geq 1$. That is, for each $n \geq 1$, μ^n is the product measure generated by μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Then for each $n \geq 1$, μ^n is a probability on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for $A \in \mathcal{B}(\mathbb{R}^n)$. Hence, as in part (i), we can construct a sequence of random variables $\{X_n : n \geq 1\}$ on some probability space $(\Omega, \mathcal{F}, P)$ such that joint distribution of $(X_1, \dots, X_n)$ is given by μ^n , which is the product measure generated by μ . Hence $\{X_n : n \geq 1\}$ is an i.i.d. sequence.

(i) WTS $X_n \rightarrow X$ a.s. (P). We need to show that
 $\exists$ some set $B \in \mathcal{B}([0,1])$ s.t. $m(B) = 0$ and

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \forall \omega \in B^c.$$

Consider that

$$X_n(\omega) = \begin{cases} 0 & \text{when } \omega = 0 \\ n & \text{when } \omega \in (0, \frac{1}{n^2}) \\ 0 & \text{when } \omega \in [\frac{1}{n^2}, 1] \end{cases}$$

Thus,

$$X_n(1) = X(\omega) = 0 \quad \forall n \geq 1$$

$$X_n(0) = X(\omega) = 0 \quad \forall n \geq 1$$

and $\forall \omega \in (0,1)$, $\exists$ some $N \in \mathbb{N}$ s.t. $\frac{1}{N^2} \leq \omega$ so that

$$X_n(\omega) = 0 \quad \forall n \geq N.$$

Thus, $B = \emptyset$ and $m(\emptyset) = 0$, and

$$\lim_{n \rightarrow \infty} X_n(\omega) = X_n(\omega) = 0 \quad \forall \omega \in B^c = [0,1].$$

Thus, $X_n \rightarrow X$ a.s. (P).

(ii) Want to determine if $X_n \rightarrow X$ in probability. That is we want to show that $\forall \varepsilon > 0$

$$\lim_{n \rightarrow \infty} m(\{|X_n - X| > \varepsilon\}) = 0.$$

Consider the following argument...

$$\lim_{n \rightarrow \infty} m(\{\omega: |X_n(\omega) - X(\omega)| > \varepsilon\})$$

$$= \lim_{n \rightarrow \infty} m(\{\omega: \omega \in (0, \frac{1}{n^2})\})$$

$$= \lim_{n \rightarrow \infty} m(0, \frac{1}{n^2})$$

$$= \lim_{n \rightarrow \infty} (\frac{1}{n^2} - 0)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2}$$

$$= 0$$

Hence, $X_n(\omega) \rightarrow X(\omega)$ in probability. ✓

(iii) WTS $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$.

First, let $0 < p < 2$ ($p=1$).

We want to show that $\int |X_n|^p dm < \infty \quad \forall n \geq 1$,

$\int |X|^p dm < \infty$, and $\lim_{n \rightarrow \infty} \int |X_n - X|^p dm = 0$.

Consider that

$$\int |X_n| dm = \int n \cdot I_{(0, \frac{1}{n^2})} dm = n \cdot m(0, \frac{1}{n^2})$$

$$= n \cdot (\frac{1}{n^2} - 0) = \frac{n}{n^2} = \frac{1}{n} < \infty \quad \forall n \geq 1$$

$$\int |X| dm = \int 0 dm = 0$$

$$\lim_{n \rightarrow \infty} \int |X_n - X| dm = \lim_{n \rightarrow \infty} \int |X_n| dm = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Hence, $X_n \rightarrow X$ in L^1 .

Now let $2 \leq p < \infty$.

Consider that

$$\int |X_n|^p d\mu = \int |n \cdot I_{(0, \frac{1}{n^2})}|^p d\mu$$

$$= \int n^p I_{(0, \frac{1}{n^2})} d\mu$$

$$= n^p \cdot \mu\left(0, \frac{1}{n^2}\right)$$

$$= n^p \cdot \left(\frac{1}{n^2} - 0\right)$$

$$= \frac{n^p}{n^2}$$

$$= n^{p-2} < \infty \quad \forall n \geq 1$$

$$\int |X|^p d\mu = \int 0^p d\mu = 0$$

$$\lim_{n \rightarrow \infty} \int |X_n - X|^p d\mu = \lim_{n \rightarrow \infty} \int |X_n|^p d\mu$$

$$= \lim_{n \rightarrow \infty} n^{p-2}$$

$$> 0 \quad \forall 2 \leq p < \infty$$

Hence, $X_n \not\rightarrow X$ in L^p for $2 \leq p < \infty$. ✓

② (a) X & Y are independent and $E[X^2] < \infty$, $E[Y^2] < \infty$.

We want to show that $\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$.

$$\text{Var}[X+Y] = E[(X+Y) - E[X+Y]]^2$$

Consider that

$$E[X+Y] = \int X+Y dP_{X,Y} = \int X dP_{X,Y} + \int Y dP_{X,Y} = \iint X dP_X dP_Y + \iint Y dP_X dP_Y$$

by Fubini's since $\rightarrow$

$$= \int X dP_X + \int Y dP_Y$$

Thus, $E[X+Y] = E[X] + E[Y]$. Then

$$\text{Var}[X+Y] = E[(X+Y - E[X+Y])^2]$$

$$= E[(X+Y - (E[X] + E[Y]))^2]$$

$$= E[(X+Y - E[X] - E[Y])^2]$$

$$= E[(X - E[X]) + (Y - E[Y])]^2$$

$$= E[(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])]$$

$$= \int ((X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])) dP_{XY}$$

$$= \int (X - E[X])^2 dP_{XY} + \int (Y - E[Y])^2 dP_{XY} + 2 \int (X - E[X])(Y - E[Y]) dP_{XY}$$

$$= \iint (X - E[X])^2 dP_X dP_Y + \iint (Y - E[Y])^2 dP_Y dP_X$$

$$+ 2 \int (XY - XE[Y] - E[X]Y + E[X]E[Y]) dP_{XY}$$

$$= \int (X - E[X])^2 dP_X + \int (Y - E[Y])^2 dP_Y$$

$$+ 2 \left[\int XY dP_{XY} - E[Y] \int X dP_{XY} - E[X] \int Y dP_{XY} + E[X]E[Y] \right]$$

$$= \text{Var}[X] + \text{Var}[Y] + 2(E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y])$$

$$= \text{Var}[X] + \text{Var}[Y] + 2(E[XY] - E[X]E[Y])$$

Consider $E[XY]$ now.

$$E[XY] = \int XY dP_{XY}$$

$$= \iint XY dP_X dP_Y \quad \text{by Fubini's thm since} \\ \int |XY| dP_{XY} \leq \left(\int |X|^2 dP_{XY} \right)^{1/2} \left(\int |Y|^2 dP_{XY} \right)^{1/2} < \infty$$

$$= \int Y \int X dP_X dP_Y$$

$$= \int Y dP_Y \int X dP_X$$

$$= E[X] E[Y]$$

Then

$$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2(E[XY] - E[X]E[Y])$$

$$= \text{Var}[X] + \text{Var}[Y] + 2(E[X]E[Y] - E[X]E[Y])$$

$$= \text{Var}[X] + \text{Var}[Y] + 0$$

$$= \text{Var}[X] + \text{Var}[Y] \quad \checkmark$$

(b) Want to show $\sqrt{1 + (E[X])^2} \leq E[\sqrt{1+X^2}] \leq 1 + E[|X|]$

Let $\phi(x) = \sqrt{1+x^2}$. Then $\phi(x)$ is a convex function, and suppose that $\int |X| dP_X < \infty$ and $\int \sqrt{1+X^2} dP_X < \infty$. Thus, we can apply Jensen's inequality.

$$\phi\left(\int X dP_X\right) \leq \int \phi(X) dP_X$$

$$\Rightarrow \sqrt{1 + \left(\int X dP\right)^2} \leq \int \sqrt{1+X^2} dP_X$$

$$\Rightarrow \sqrt{1 + (E[X])^2} \leq E[\sqrt{1+X^2}] \quad \checkmark$$

If $\int |X| dP_X = \infty$ and $\int \sqrt{1+X^2} dP_X = \infty$, then the inequality is already true. Hence, this proves the first inequality.

Now, we want to show $E[\sqrt{1+X^2}] \leq 1 + E[|X|]$. Consider that

$$1+X^2 \leq 1 + 2|X| + |X|^2 = (1+|X|)^2$$

$$\Rightarrow \sqrt{1+X^2} \leq \sqrt{(1+|X|)^2}$$

$$\Rightarrow \sqrt{1+X^2} \leq 1+|X|$$

$$\Rightarrow \int \sqrt{1+X^2} dP_X \leq \int (1+|X|) dP_X$$

$$\Rightarrow E[\sqrt{1+X^2}] \leq 1 + \int |X| dP_X$$

$$\Rightarrow E[\sqrt{1+X^2}] \leq 1 + E[|X|] \quad \checkmark$$

Hence, the second inequality has been proved. $\checkmark$

(c) We have $P_X \ll m$ and $P_Y \ll m$ with

$$\frac{dP_X}{dm} = f \quad \text{and} \quad \frac{dP_Y}{dm} = g.$$

We want to show that $P_{(X,Y)} \leq m^2$ and

$$\frac{dP_{(X,Y)}}{dm^2} = f \cdot g \quad \text{a.e. } (m^2).$$

Since $P_X \ll m$ and $\frac{dP_X}{dm} = f$, we have (by the R-N thm)

$$P_X(A_1) = \int_{A_1} \frac{dP_X}{dm}(x) dm(x) = \int_{A_1} f(x) dm(x) \quad \text{where } A_1 \in \mathcal{B}(\mathbb{R}).$$

Since $P_Y \ll m$ and $\frac{dP_Y}{dm} = g$, we have (by the R-N thm)

$$P_Y(A_2) = \int_{A_2} \frac{dP_Y}{dm}(y) dm(y) = \int_{A_2} g(y) dm(y)$$

where $A_2 \in \mathcal{B}(\mathbb{R})$.

Then for $A = A_1 \times A_2 \in \mathcal{B}(\mathbb{R}^2)$

$$P_{(X,Y)}(A) = P_X(A_1) \times P_Y(A_2) \quad (\text{since } X \text{ and } Y \text{ are independent})$$

$$= \int_{A_1} f(x) dm(x) \cdot \int_{A_2} g(y) dm(y) \quad (\text{as we defined before})$$

$$= \int_{A_2} \int_{A_1} f(x) g(y) dm(x) dm(y) \quad \left(\begin{array}{l} \text{by Tonelli's thm since} \\ f \text{ and } g \text{ are non-negative} \end{array} \right)$$

$$= \int_{A_1 \times A_2} f(x) g(y) dm^2(x, y)$$

Then, since we can write $P_{(X,Y)}$ in this form where $f \cdot g$ is a non-negative function (since f and g are both non-negative functions), we have that

$$P_{(X,Y)} \ll m^2$$

and by the Radon-Nikodym derivative

$$\frac{dP_{(X,Y)}}{dm^2} = f \cdot g$$

a.e. (m^2) (by the uniqueness of the R-N derivative).

(d) Suppose that f is continuous and bounded ^{on $\mathbb{R}$} . Then we want to show that $f * g$ is also continuous and bounded.

Let us first show that $f * g$ is bounded. Since f is bounded, suppose that $M \in \mathbb{R}_+$ s.t. $f(x) \leq M \quad \forall x \in \mathbb{R}$. Then

$$\begin{aligned} f * g(x) &= \int f(x-u)g(u)dm(u) \\ &\leq \left| \int f(x-u)g(u)dm(u) \right| \\ &\leq \left| \int M g(u)dm(u) \right| \\ &= M \left| \int g(u)dm(u) \right| \\ &= M \cdot \left| \int g(u)dm(u) \right| \\ &\leq M \cdot \int |g(u)|dm(u) \\ &< \infty \end{aligned}$$

* This is true for all $x \in \mathbb{R}$.

(since we know $g \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$).

Thus, $f * g(u)$ is bounded by the function $M \cdot \int |g(u)|dm(u)$.

Now, let us show that $f * g(u)$ is continuous. Consider a sequence $\{x_n\}_{n \geq 1}$ such that $x_n \rightarrow x$ as $n \rightarrow \infty$. Then consider that the function $h(x) = x - u$ ^{where $u \in \mathbb{R}$} is a continuous function, so $h(x_n) \rightarrow h(x)$ as $n \rightarrow \infty$. Furthermore, we have that f is a continuous function, so

$$f(h(x_n)) = f(x_n - u) \rightarrow f(x - u) \quad \text{as } n \rightarrow \infty.$$

Finally, let $g(u)$ be a constant (for a particular value of u). Then $f(x_n - u)g(u) \rightarrow f(x - u)g(u)$ as $n \rightarrow \infty$.

Now, we can apply the DCT. That is, because

(i) $|f(x_n - u)g(u)| \leq M \cdot |g(u)| \quad \forall x \in \mathbb{R}$ since f is bounded, so that $f(x - u)g(u)$ is bounded for all $n \geq 1$,

(ii) $\int |g(u)| du < \infty$ since we are told that $g \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$, and

(iii) $f(x_n - u)g(u) \rightarrow f(x - u)g(u)$ as $n \rightarrow \infty$ (which we just showed),

the DCT implies that

$$\int f(x_n - u)g(u) dm(u) \rightarrow \int f(x - u)g(u) dm(u)$$

as $n \rightarrow \infty$. That is

$$f * g(x_n) \rightarrow f * g(x) \quad \text{as } n \rightarrow \infty,$$

so we find that $f * g$ is continuous. ✓

(e) Let $\lambda = P_X * P_Y$. WTS $\frac{d\lambda}{dm} = f * g$ a.e. (m), where

$$f = \frac{dP_X}{dm} \quad \text{and} \quad g = \frac{dP_Y}{dm} \quad (+ dP_X \ll m, dP_Y \ll m).$$

Let us consider $\lambda(A)$ where $A \in \mathcal{B}(\mathbb{R}^2)$.

$$\lambda(A) = (P_X * P_Y)(A) \quad (\text{by definition})$$

$$= \iint I_A(x+y) dP_X(x) dP_Y(y) \quad (\text{by def. of conv. of measures})$$

$$= \iint I_A(x+y) f(x) dm(x) dP_Y(y) \quad (\text{since } dP_X = f \cdot dm)$$

$$= \iint I_A(u) f(u-y) dm(u) dP_Y(y) \quad \text{where } u = x+y \text{ and making use of change of variables}$$

$$\begin{aligned}
&= \iint \mathbb{I}_A(u) f(u-y) d\Pr(y) dm(u) \quad \left(\text{by Fubini's since } f \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m) \right) \\
&= \int \mathbb{I}_A(u) \int f(u-y) d\Pr(y) dm(u) \\
&= \int_A \int f(u-y) g(y) dm(y) dm(u) \quad (\text{since } d\Pr = g \cdot dm) \\
&= \int_A f * g(u) dm(u) \quad (\text{by def. of the conv. of 2 functions})
\end{aligned}$$

Consider that f and g are both non-negative functions, so $f * g$ is a non-negative function. Since we can write $\lambda(A)$ in this way, we get that $\lambda \ll m$ and by the definition of the Radon-Nikodym

$$f * g(u) = \frac{d\lambda}{dm}(u)$$

a.e. (m) (by the uniqueness of the Radon-Nikodym derivative).

(f) If Z is a random variable with $P_Z = \lambda$ (as in the last problem), we want to determine how Z is related to X and Y .

With $\lambda = P_X * P_Y = P_Z$ and X and Y are independent, we will have

$$Z = X + Y.$$

Furthermore, Z will have density of $f * g$.
That is, the law of Z will be $f * g$.

③ (i) WTS that a normal random variable exists...
 Let $F(x) = \Phi(x)$ (as defined on the exam). This is a non-decreasing, right-continuous function. Then define

$$\mathcal{C} = \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\},$$

and define μ_Φ as

$$\mu_\Phi((a, b]) = \Phi(b+) - \Phi(a+)$$

$$\mu_\Phi((a, \infty)) = \Phi(\infty) - \Phi(a+)$$

where $\Phi(x+) = \lim_{y \downarrow x} \Phi(y)$ and $\Phi(\infty) = \lim_{x \uparrow \infty} \Phi(x)$. Then it can be shown that:

(i) $\mathcal{C}$ is a semi-algebra

(ii) μ_Φ is a measure on $\mathcal{C}$.

Furthermore, for $A \in \mathcal{R}$, let

$$\mu_\Phi^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu_\Phi(A_n) : \{A_n\}_{n=1}^{\infty} \subset \mathcal{C}, A \subset \bigcup_{n=1}^{\infty} A_n \right\}.$$

Then by Caratheodory's extension theorem:

(i) μ^* is an outer measure

(ii) $\mathcal{C} \subset \mathcal{M}_{\mu_\Phi^*}$ where $\mathcal{M}_{\mu_\Phi^*} = \{A : A \text{ is } \mu^*\text{-measurable}\}$

and by another result we have that

(1) $\mathcal{M}_{\mu_\Phi^*}$ is a σ -algebra,

(2) μ_Φ^* restricted to $\mathcal{M}_{\mu_\Phi^*}$ is a measure, and

(3) $\mu^*(A) = 0 \Rightarrow \mathcal{P}(A) \subset \mathcal{M}_{\mu_\Phi^*}$ (which means that $(\Omega, \mathcal{M}_{\mu_\Phi^*}, \mu_\Phi^*)$ is a complete measure space).

(iii) $\mu_{\Phi}^* = \mu_{\Phi}$ on $\mathcal{C}$.

← Carathéodory extension of μ_{Φ}

We have shown that $(\mathbb{R}, M_{\mu_{\Phi}^*}, \mu_{\Phi}^*)$ is a measure space. Consider that $\mathcal{C} \subset M_{\mu_{\Phi}^*}$, and we know that

$$\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}).$$

Since $M_{\mu_{\Phi}^*}$ is a σ -algebra, $\mathcal{C} \subset M_{\mu_{\Phi}^*} \Rightarrow \mathcal{B}(\mathbb{R}) \subset M_{\mu_{\Phi}^*}$, so $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_{\Phi}^*)$ is also a measure space. We also know that μ_{Φ}^* restricted to $\mathcal{B}(\mathbb{R})$ will equal μ_{Φ} .

Thus, μ_{Φ}^* is an extension of μ_{Φ} on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. (Note that μ_{Φ}^* is unique by the uniqueness of measures theorem.)

Thus, let $(\Omega, \mathcal{A}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_{\Phi}^*)$ be a probability space, and let $X: \mathbb{R} \rightarrow \mathbb{R}$ be a function s.t.

$$X(\omega) = \omega \quad \forall \omega \in \mathbb{R}.$$

Then X is a meas. function since it is a constant, and $\mu_{\Phi}^*(\mathbb{R}) = \mu_{\Phi}(\mathbb{R}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = 1$, so μ_{Φ}^* is a probability measure. Thus, X is a random variable!

Note that

$$F_X(x) = P(X \leq x) = \mu_{\Phi}^*((-\infty, x]) = \Phi(x) \quad \forall x \in \mathbb{R}$$

and

$$P_X(A) = P(X^{-1}(A)) = \mu_{\Phi}^*(A) \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

Hence, a normal(0,1) random variable exists!

(ii) Want to compute the Lebesgue integral

$$\int_{\Omega} X(x) dP(x).$$

Consider the following...

$$\int_{\Omega} X(x) dP(x) = \int_{\Omega} I(X(x)) dP(x) \quad \text{where } I(x) = x$$

$$= \int_{\mathbb{R}} I(x) dP(X^{-1}(x)) \quad \text{by change of variables}$$

$$= \int_{\mathbb{R}} x dP_x(x) \quad \text{by def. of } P_x \rightarrow \text{the measure induced by } X$$

$$= \int_{\mathbb{R}} x \phi(x) dm(x) \quad \text{since } \frac{dP_x}{dm} = \phi$$

$$= \int_{\mathbb{R}} x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dm(x) \quad \Rightarrow dP_x = \phi dm$$

$$= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= 0$$

(iii) For $n \geq 1$, let μ_n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$.

We want to determine under what conditions can we say that there is a probability measure P on $(\mathbb{R}^{\infty}, \mathcal{B}(\mathbb{R}^{\infty}))$ s.t. $P(A \times \mathbb{R}^{\infty}) = \mu_n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$ for $n \geq 1$.

This is the case when for all $n \geq 1$, $\mu_{n+1}(B \times \mathbb{R}) = \mu_n(B) \quad \forall B \in \mathcal{B}(\mathbb{R}^n)$. $\rightarrow$ by Kolmogorov's consistency thm $\rightarrow$

We also want to describe a sequence of random variables $\{X_n: n \geq 1\}$ defined on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty), P)$ s.t. the joint distribution on $(X_1, \dots, X_n)$ is given by $\mu_n \forall n \geq 1$.

Let $X_n(\omega)$ for $\omega = (\omega_1, \omega_2, \dots) \in \mathbb{R}^\infty$ be defined as $X_n(\omega) = \omega_n$. Then

$$P_{(X_1, \dots, X_n)}(A) = \mu_n(A) \quad \forall A \in \mathcal{B}(\mathbb{R}^n) \text{ and } n \geq 1.$$

(iv) Now, let us show that there exists a sequence of iid $N(0, 1)$ random variables: $\{X_n: n \geq 1\}$.

Define μ_F where $F = \Phi$ is the $N(0, 1)$ CDF. Then let

$$\mu_1 = \mu_F \text{ on } (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$\mu_2 = \mu_1 \times \mu_F \text{ on } (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$$

⋮

$$\mu_n = \mu_{n-1} \times \mu_F \text{ on } (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$$

(We define this with cross products, because we want independence here.)

Note that the consistency conditions hold:

$$(i) \mu_n(\mathbb{R}^n) = \underbrace{\mu_F(\mathbb{R}) \times \dots \times \mu_F(\mathbb{R})}_n = 1 \times \dots \times 1 = 1$$

Thus, μ_n is a probability measure.

$$\begin{aligned} (ii) \quad \mu_n(B \times \mathbb{R}) &= \mu_n(B) \times \mu_F(\mathbb{R}) \\ &= \mu_n(B) \times 1 \\ &= \mu_n(B) \end{aligned}$$

for $B \in \mathcal{B}(\mathbb{R}^n)$ and $n \geq 1$. ✓

Thus, Kolmogorov's consistency theorem tells us that $\exists$ a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $\forall n \geq 1$,

$$P_{(X_1, \dots, X_n)}(B) = \mu_n(B) = \underbrace{\mu_F(B_1) \times \dots \times \mu_F(B_n)}_n$$

Thus, the joint distribution of $X_1, \dots, X_n$ is

$$\mu_n = \underbrace{\mu_F \times \dots \times \mu_F}_n,$$

so $\{X_n; n \geq 1\}$ is a sequence of iid $N(0, 1)$ random variables.